

16.6 Answers

- ① P does not lie on the surface
 Q does lie on the surface

② a) $4x - y - z = -4$; a plane

b) $z = y^2 - x^2$; hyperbolic paraboloid

c) $y = x^2 + z^2$; paraboloid (circular)

③ a) $\vec{r}(u, v) = \langle u, v, 2u^2 + v^2 \rangle$

b) $\vec{r}(u, v) = \langle u, v, -\sqrt{1 - 2u^2 - 4v^2} \rangle$

c) $\vec{r}(u, v) = \langle u, \frac{\sqrt{1 - 2u^2 - v^2}}{2}, v \rangle$

d) let $x = 2 \sin \phi \cos \theta$, $y = 2 \sin \phi \sin \theta$,
 $z = 2 \cos \phi$, where $0 \leq \phi \leq \frac{\pi}{4}$
and $0 \leq \theta \leq 2\pi$.

④ $-x + 2z = 1$

⑤ a) $3\sqrt{14}$, b) $\frac{2\pi}{3}(2\sqrt{2} - 1)$, c) $\frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$

d) $\frac{\pi}{2} [\sqrt{2} + \ln(1 + \sqrt{2})]$ (you may want to
use an integral
table here)

e) 4